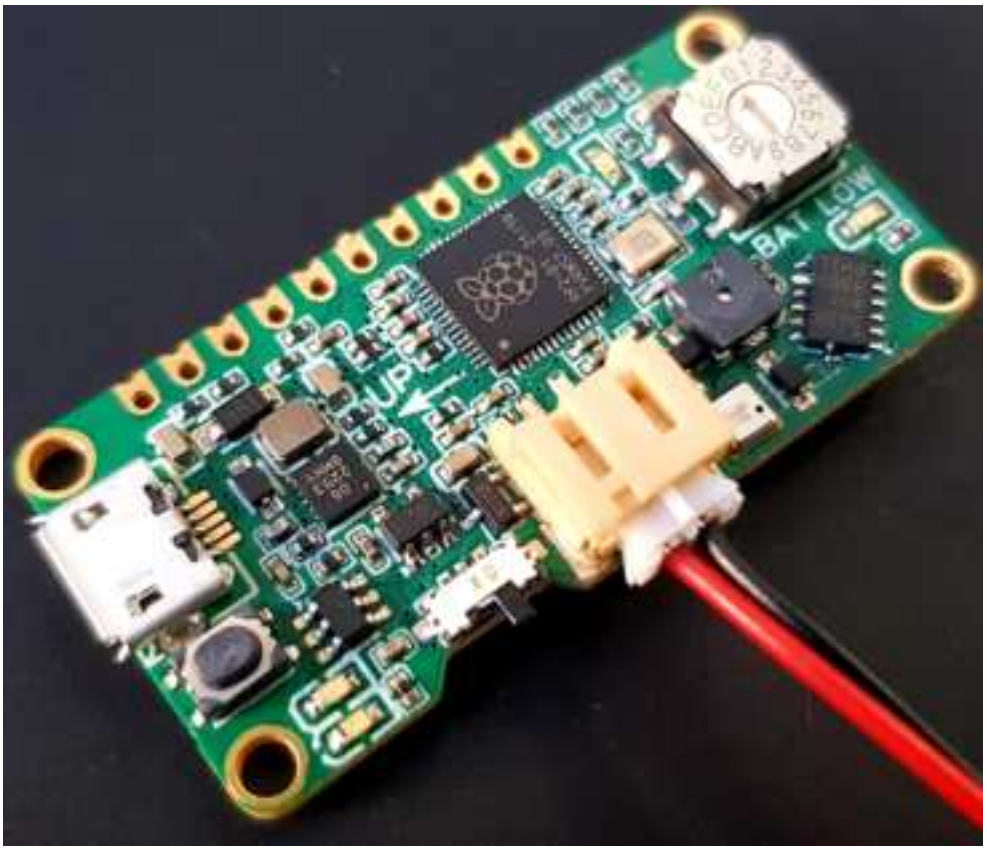
Rockit Rev. 1

User Manual

Firmware 3.0



Triskel Instruments

formerly DanInvents

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1 About this manual

This manual explains how to prepare, configure, fly and update the Rockit Rev. 1 flight computer using Firmware 3.0.

Rockit was originally released under the name **DanInvents**. The product is now developed and supported by **Triskel Instruments**. Older manuals or videos may still carry the DanInvents name.

Firmware 3.0 is a major software update. It introduces a new MicroPython-based firmware, improved launch and apogee detection, safer recovery sequencing, persistent settings, and a standalone Windows updater.

2 Disclaimer

In no respect shall Triskel Instruments be accountable for liabilities, claims, demands, damages or suits resulting from the use of Rockit and/or its related firmware.

By using this product, you accept the risks associated with model rocketry, lithium batteries, servo-driven recovery systems and experimental electronics.

Always test your complete recovery system on the ground before flight.

3 What is Rockit?

Rockit, or *Rocket Operation Computing Kit*, is a compact flight computer designed for model rocketry.

It provides:

- altitude and acceleration measurement;
- automatic or timed recovery deployment;
- control of two servo motors;
- flight-data logging to a microSD card;
- maximum-altitude reporting after flight;
- battery charging for a single-cell lithium battery.

3.1 Main specifications

Specification	Value
Dimensions	44 mm × 22 mm × 9 mm
Mass	5 g
Input voltage	2.3 to 5.6 V
Idle current	Approx. 70 mA
Logging current	Approx. 80 mA
Peak current while beeping	Approx. 200 mA
Typical current while moving a servo	Approx. 400 mA
Maximum continuous 5 V output current	2 A
Battery charging current	500 mA
Sampling rate	100 Hz
Altimeter range	Up to 9160 m
Accelerometer range	Up to 22.6 g

The practical lower voltage limit is often set by the protection circuit inside the LiPo battery. Many protected batteries disconnect themselves at around 3 V.

4 Board layout

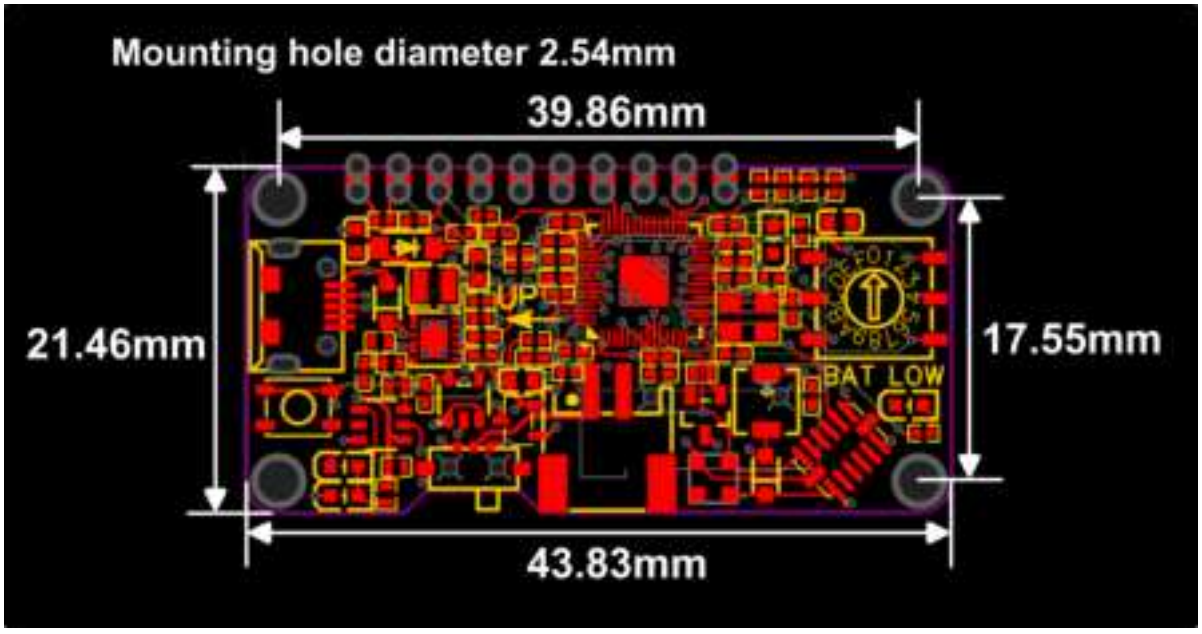


Figure 1: Rockit Rev. 1 dimensions. The 2.54 mm holes are connected to ground and can be used for probing.

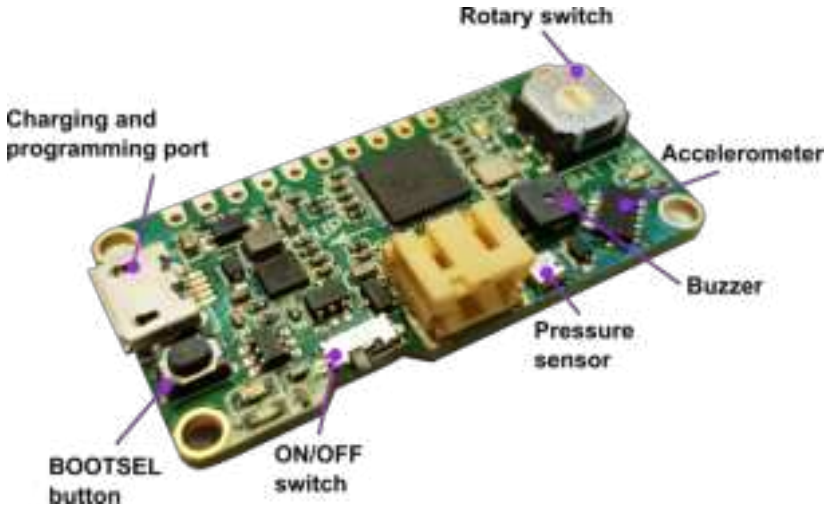


Figure 2: Top-side layout. The I^{2}C pins and servo signal pins use 3.3 V logic.

Do not apply 5 V logic to the 3.3 V signal pins. Doing so may damage the RP2040 microcontroller.

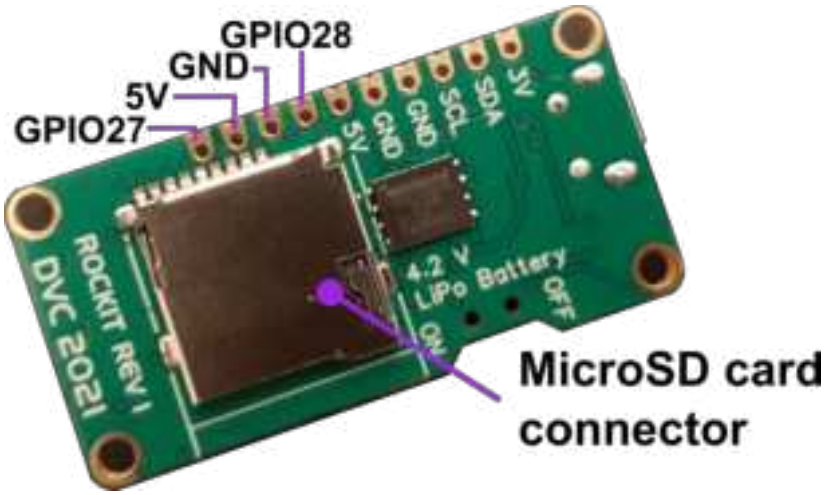


Figure 3: Bottom-side layout. Servo 1 uses GPIO 28 and Servo 2 uses GPIO 27. I^{2}C SDA and SCL use GPIO 10 and GPIO 11.

5 Before the first flight

Before flying Rockit, check the following:

1. Insert a compatible microSD card.
2. Connect the battery with the correct polarity.
3. Configure both servo start and end positions.
4. Configure the required deployment delays.
5. Run the position-3 recovery-system test.
6. Confirm that the battery can move both servos without shutting down.

7. Make sure the pressure sensor is protected from direct sunlight but its vent hole is not blocked.

6 Power and battery

Rockit is designed to operate from 2.3 to 5.6 V.

The onboard battery charger is intended only for a **single-cell (1S) lithium-ion or lithium-polymer battery**, typically 3.7 V nominal.

Battery polarity is critical. If Rockit becomes unusually hot immediately after connecting the battery, disconnect it at once.

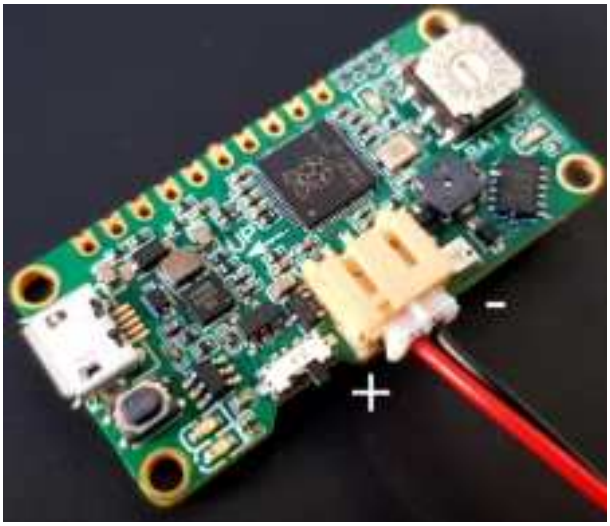


Figure 4: Correct JST battery polarity. Red is battery positive and black is ground.

Servo motors can draw large current spikes. A battery with an aggressive protection circuit may shut itself off when a servo moves.

Always test the complete battery and servo combination before flight.

7 Choosing a microSD card

For best compatibility:

- use a Class 10 microSD card;
- use 16 GB or less when possible;
- format larger cards as FAT32 if necessary.

If Rockit appears to start but does not proceed normally, the microSD card is one of the first things to check.

8 Rotary switch quick reference

The rotary switch is used both to select the flight mode and to configure Rockit.

Position	Function
A	Automatic mode: deploy at detected apogee
B	Timer mode: deploy after a preset time from launch detection
C	Set Servo 1 delay after apogee in Automatic mode
D	Set Servo 1 delay after launch in Timer mode
E	Set Servo 1 start position
F	Set Servo 1 final position
0	Set Servo 2 start position
1	Set Servo 2 final position
2	Set additional delay before Servo 2 moves
3	Test the complete servo/recovery sequence
4–9	Inactive / non-flight positions

Some Rockit boards use a standard-coded rotary switch and some use a complementary-coded switch. Firmware 3.0 supports both. The Windows updater can detect the correct type automatically.

9 Changing settings

The basic configuration workflow is:

1. Turn Rockit off.
2. Turn the rotary switch to the setting you want to change.
3. Turn Rockit on.
4. Follow the LED/buzzer feedback.
5. Rotate the switch to choose the desired value.
6. Keep the value stable so Rockit can save it.
7. Turn Rockit off when the setting has been stored.

Firmware 3.0 stores the settings in a persistent configuration file, so they remain available after power is removed and can also be backed up by the firmware updater.

9.1 Setting the deployment delays

There are three timing settings:

- **Position C**: delay between apogee detection and Servo 1 movement in Automatic mode;
- **Position D**: delay between launch detection and Servo 1 movement in Timer mode;
- **Position 2**: additional delay before Servo 2 moves.

Automatic mode delay – position C

Each rotary step adds **0.5 seconds**.

Switch value	0	1	2	3	4	5	10	15
Delay (s)	0	0.5	1.0	1.5	2.0	2.5	5.0	7.5

Timer mode delay – position D

Each rotary step adds **1 second**.

Switch value	0	1	2	3	4	5	10	15
Delay (s)	0	1	2	3	4	5	10	15

Servo 2 additional delay – position 2

Each rotary step adds **0.5 seconds** after Servo 1.

Switch value	0	1	2	3	4	5	10	15
Additional delay (s)	0	0.5	1.0	1.5	2.0	2.5	5.0	7.5

For Servo 2, think of the setting as "how long after Servo 1 should Servo 2 move?" This is usually easier to understand than calculating the total time from launch or apogee.

9.2 Setting the servo positions

Use:

- **E** for Servo 1 start position;
- **F** for Servo 1 final position;
- **0** for Servo 2 start position;
- **1** for Servo 2 final position.

When adjusting a servo, the rotary switch gives 16 possible positions from 0 to F.

You do not need to calculate an angle. Simply choose the position that places your mechanism where you want it.

If a servo does not appear to move during deployment, check that its start and final positions are not set to the same value.

10 Testing the recovery system

Position **3** performs a complete ground test of the recovery servos.

With Rockit turned off:

1. turn the rotary switch to **3**;
2. power Rockit on;
3. Servo 1 moves to its programmed start position;
4. Servo 2 moves to its programmed start position;
5. Rockit waits approximately 5 seconds;

6. Servo 1 moves to its programmed final position;
7. Servo 2 moves to its programmed final position;
8. the flight computer then remains inactive.

Use position 3 before every flight in which you have changed the recovery mechanism, servo linkage, battery, or servo settings.

11 Flight modes

11.1 Automatic mode – position A

Automatic mode is intended for normal recovery deployment at apogee.

After startup:

1. Rockit arms after approximately 3 seconds.
2. Rockit waits for launch.
3. Launch is confirmed using both altitude and acceleration information.
4. During ascent, Rockit tracks the filtered altitude.
5. After apogee is confirmed, Rockit waits for the delay configured with position C.
6. Servo 1 moves.
7. Servo 2 moves after the additional delay configured with position 2.

11.2 How launch detection works

Rockit does not trigger from a single bump or a single pressure change.

For a real flight, the firmware looks for:

- an altitude increase of about **10 m**;
- a change in acceleration of more than about **2 g**;
- the acceleration condition to persist for multiple samples;
- both events to occur close together in time.

This makes false launch detection from handling or vibration much less likely than in earlier firmware versions.

11.3 How apogee detection works

Firmware 3.0 no longer declares apogee just because one or two altitude samples decrease.

Instead, it:

- tracks the highest filtered altitude reached;
- ignores very early false peaks immediately after launch;

- waits until the rocket has climbed to a meaningful altitude;
- requires a clear drop from the peak;
- requires that drop to persist briefly before deployment begins.

Rockit waits for a clear and persistent descent from the detected altitude peak before treating the rocket as being past apogee.

11.4 Timer mode – position B

Timer mode deploys the recovery system after a fixed time from launch detection.

After launch is confirmed:

1. Rockit waits for the time configured with position D.
2. Servo 1 moves.
3. Rockit waits for the additional Servo 2 delay configured with position 2.
4. Servo 2 moves.

Timer mode is useful when you want predictable deployment timing rather than apogee-based deployment.

12 What happens after launch?

During flight, Rockit records data to the microSD card.

The logged data include:

- time;
- altitude;
- filtered altitude;
- longitudinal acceleration;
- perpendicular acceleration;
- temperature.

Rockit also keeps a short pre-launch buffer, so the saved file contains data from immediately before takeoff.

Approximately one minute after launch, Rockit transitions to maximum-altitude reporting.

13 Reading the maximum altitude

After the flight, Rockit repeatedly reports the maximum altitude using the buzzer.

The pattern is:

1. two short beeps mark the start;
2. each altitude digit is beeped separately;

3. two short beeps mark the end;
4. the pattern repeats.

The digit **0** is represented by a long beep.

Example: 67 m

You will hear:

two short beeps → six short beeps → seven short beeps → two short beeps

The complete flight record can also be read directly from the CSV file stored on the microSD card.

14 Updating Rockit

Firmware 3.0 introduces a standalone Windows updater.

You no longer need Python, Arduino IDE, VS Code or `mpremote`.

14.1 Recommended update method

1. Download the latest Rockit firmware updater from:

   https://github.com/DanInvents/Rockit/releases

2. Connect Rockit to the computer using USB.
3. Run `Rockit_Firmware_Updater.exe`.
4. Follow the instructions shown on screen.
5. If asked, turn the rotary switch to position A so the updater can identify the switch coding.
6. Wait until the updater displays **UPDATE SUCCESSFUL**.

The updater can:

- identify an existing MicroPython installation;
- back up saved Rockit settings;
- install Firmware 3.0;
- restore servo positions and recovery delays;
- detect standard or complementary-coded rotary switches;
- guide older Arduino-based Rockit units through BOOTSEL migration;
- install the validated MicroPython runtime automatically.

If you are updating an older Arduino-based Rockit, its old EEPROM settings cannot be transferred automatically. Configure and test the servos again after the migration.

15 What changed in Firmware 3.0?

Firmware 3.0 is the largest software revision since Rockit was introduced.

The most important improvements are:

- more reliable launch detection using altitude and acceleration together;
- more robust automatic apogee detection;
- improved recovery-servo timing and sequencing;
- improved servo startup and detach behavior;
- a dedicated position-3 recovery-system test;
- a shorter arming delay;
- persistent human-readable settings;
- settings backup and restore during firmware updates;
- support for both rotary-switch hardware variants;
- improved pre-launch buffering and SD-card logging;
- clearer maximum-altitude reporting;
- a much faster post-flight report;
- improved buzzer and PWM handling;
- standalone Windows firmware updates;
- migration from the previous Arduino firmware to MicroPython.

Firmware 3.0 has been flight-tested successfully in both:

- **A – Automatic mode**;
- **B – Timer mode**.

16 Frequently asked questions

My flight computer gets very hot when I connect the battery. What should I do?

Disconnect the battery immediately. The battery connector polarity is probably reversed.

Do not leave the battery connected while diagnosing the problem.

My servo does not move.

Check the following:

- the servo is connected to the correct output;
- the battery can supply enough current;
- the start and final positions are different;

- the mechanism is not mechanically jammed;
- the position-3 ground test works correctly.

Rockit does not start normally.

Check the microSD card first.

For best compatibility, use a Class 10 card of 16 GB or less.

Also check that the battery voltage is adequate.

Can I test launch detection by tilting the board?

No. Firmware 3.0 uses both acceleration and altitude changes to detect launch.

For checking the recovery mechanism, use rotary position **3** instead of trying to simulate a complete launch.

What is the difference between standard and complementary-coded Rockit boards?

Some production batches used rotary switches with opposite electrical coding.

Firmware 3.0 supports both types using the same firmware.

The Windows updater can detect the correct type automatically by asking you to place the switch at position **A**.

Do I need Arduino IDE or Python to update Rockit?

No.

For Firmware 3.0, use the standalone Windows updater from the Rockit GitHub Releases page.

17 Support and firmware downloads

Firmware releases and update information are available at:

https://github.com/DanInvents/Rockit/releases

Rockit is developed and supported by:

Triskel Instruments
formerly DanInvents